Navigating Privacy Policies with NLP and Graph Mining: Advancements in User-Centric Legal Document Analysis

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***Abstract*—As mobile, healthcare, shopping, gaming, and other sectors become increasingly popular, users are frequently required to consent to privacy policies that explain how their personal information will be collected, used, and shared. However, these policies are often lengthy and complex, making it difficult for users to fully understand them, which leaves many users unaware of potential risks to their privacy and data security. This research paper addresses the growing need for privacy-conscious software across various sectors, setting the stage for future developments in accountable and transparent system design. This article investigates the application of Natural Language Processing, graph mining, and knowledge graph analysis as transformative tools to enhance the analysis of privacy policies. This study explores how the current state-of-the-art research enhances the comprehensibility and transparency of PP documents, ultimately empowering users to understand their privacy rights and how their data may be shared. We focus on automating the summarization and simplification of PPs to make them more user-friendly. Our approach includes flagging potential issues that may help identify risks, generating simplified summaries of policies, and enabling users to make informed decisions about how their personal data is used. This initiative aims to build trust in the management of personal data.**

***Index Terms*—NLP, Privacy Policy, Knowledge Graph**

I. Introduction

In the modern digital ecosystem, Privacy policies (PPs) play a pivotal role in defining the data usage terms and protection between organizations and users. Users are frequently required to agree to PPs when using apps across various domains, including mobile, healthcare, shopping, and gaming. Privacy policies (PPs) are widespread, but they are often lengthy, opaque, and difficult to understand. Many users struggle to grasp how their personal data is collected, stored, shared, and protected. This raises concerns about customer rights and the ethical responsibilities of service providers.

As data regulations become increasingly strict and complex, their lack of accessibility hinders privacy awareness and informed decision-making. The gap between legal text and user comprehension is a major issue in privacy and transparency.

Advancements in Natural Language Processing (NLP) and graph mining offer promising solutions for simplifying and analyzing privacy policies at scale. These technologies enable automated parsing, summarization, and semantic structuring of legal documents, making them more user-centric [1]. By utilizing NLP techniques to identify important clauses within PP documents, applying sophisticated, complex network/graph-based models, and visually representing the relationships between different components of these policies, researchers and practitioners develop innovative tools to make complex policies more accessible to the general public, ultimately leading to greater user understanding and empowerment in navigating and engaging with policy matters.

This article addresses these issues by developing a framework for automated PP analysis focusing on transparency, accountability, and preserving customer rights. Our goal is to design accountable software systems that prioritize user privacy, ensuring that customers know how their data is used while maintaining compliance with data protection regulations such as the General Data Protection Regulation (GDPR), Health Insurance Portability and Accountability Act (HIPAA), and California Consumer Privacy Act (CCPA).

In their article, Cui et al., [1] proposed a knowledge graph (KG) framework called PoliGraph, designed to represent entities and their relationships within PP texts. The authors developed PoliGraph-er, an NLP-based tool that automatically extracts these graphs from policy documents. This approach helps in the analysis of privacy policies, moving beyond the traditional sentence-by-sentence examination. The study also revisits the concept of ontologies, distinguishing between local and global contexts to better capture term subsumptions within individual policies and broader legal frameworks. The Poligraph framework uses n-grams, a contiguous sequence of n words or tokens in a text to preprocess the text in the construction of knowledge graphs. N-grams represent fixed-length word sequences for tasks like text generation, language modeling, or feature extraction, which are typically used for statistical language models or frequency-based analysis [2]. The n-gram model considers only adjacent words within the n-length window for tokenizing the text. It is limited to fixed n length and cannot skip intermediate words. These simpler-

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to-compute n-gram tokenizers rely on frequency counts.

The skip-gram model is an alternative probabilistic model used om word embedding techniques (e.g., Word2Vec) to learn vector representation of words [3]. It generates word pairs from a sliding window over the text, skipping over words in between. This technique is used in neural network models for learning distributed word embeddings. It considers both adjacent and non-adjacent words, introducing more contextual diversity, and is used for Word embedding generation (e.g., Word2Vec). Skip-grams can be used for semantic analysis, which requires dense vector representations of words. Therefore, they are computationally more intensive and require a neural model or optimization algorithm. In this article, we implemented the Poligraph algorithm for KG construction using the skip-gram model instead of the traditional n-gram model to compare the significant differences in how semantic relationships between privacy policy terms are identified and represented.

We formulated the following research questions.

RQ1: How can we assess the strengths, quality, and coverage of the entities involved in the PPs?

RQ2: How does using skip-gram for constructing knowledge graphs to analyze PPs differ from n-gram implementation in PoliGraph?

RQ3: How can PPs be analyzed to identify potential risks to customer rights and personal data privacy?

RQ4: What methods can be used to design software systems that ensure transparency and accountability in data collection, usage, and sharing?

Although terms of service (TOS) agreements and PPs are essential for creating accountability and transparency in digital platforms, consumers frequently ignore them because of their length and complexity. Empirical research indicated that 74% of participants acknowledged completely ignoring PPs in favor of quick-join alternatives that circumvent these guidelines [4]. Even when consumers interact with such rules, they pay little attention to them; 98% of them were unable to recognize purposefully included "gotcha" elements, including consents to transfer their firstborn child or exchange data with government agencies. The effectiveness of PPs and the harmony between user autonomy and digital governance are seriously called into question by these findings, which highlight the widespread difficulties in user comprehension and interaction with PPs and TOS.

II. Related Work

We present an overview of the related work in the area of analyzing PPS using AI-based techniques and the efficiency of KG in representing and analyzing the structural and legal implications, as well as the advantages and limitations of these studies I.

Elluri et al. [5], (2018) developed a Knowledge Graph to illustrate the rules of PCI DSS and EU GDPR using an ontology that identifies obligations from these regulations and links them to relevant Cloud Security Alliance controls, validated it against the data policies of major Big Data vendors.

Cui et al. [1] presented PoliGraph, a knowledge graph architecture for structuring relationships and PoliGraph-ER, an NLP-based tool for extracting relationships from PPs. The benefit of this method is that it can examine PPs in-depth by extracting relational data from textual statements, which helps in better-comprehending privacy practices. However, their framework may not be as flexible for more comprehensive privacy compliance scenarios as it mostly concentrates on data collection practices, under-examining other facets of PPs.

To extract privacy-related implications from policy documents, Bhatia et al. [6] used a hybrid approach that used crowd-sourcing and NLP techniques. Their method struck a mix between automation and human judgment, achieving a great recall (0.85) and precision (0.93) in identifying privacy goals. However, the need for high-quality annotations for dependency parsing and the difficulties with full automation and scalability point to areas that need to be improved in the future, particularly for extensive PP analysis.

Martinelli et al. [7] investigated how to find sensitive information in text documents using transfer learning, unsupervised learning, and NLP. Their semi-automated method produced promising recall and precision rates while reducing manual work. Nonetheless, the approach's dependence on domain-specific knowledge representation and the ongoing requirement for expert validation point to the need for additional improvement, especially when it comes to expanding and diversifying the datasets.

By combining Named Entity Recognition (NER) models with NLP tools including NLTK, Stanford CoreNLP, and spaCy, Silva et al. [8] tackled the identification of Personally Identifiable Information (PII). With a remarkable F1 score of over 90%, their strategy encouraged adherence to privacy laws. Notwithstanding this achievement, the human labor involved in labeling datasets and the restricted scalability brought on by the scarcity of high-quality labeled datasets continue to be obstacles for practical implementation.

Carson et al. [9] used NLP techniques like TF-IDF and Bag of Words in combination with machine learning models including logistic regression, Naïve Bayes, Support Vector Machines (SVM), and neural networks to study and categorization of PPs. Their research showed that it is possible to automate the categorization of web-based policies with an accuracy up to 84%. However, because of uneven annotations and varying terminology, their study showed lower accuracy (67%) when applied to IoT privacy standards, highlighting the need for bigger, more diversified datasets and stronger models.

In order to identify GDPR violations, Alshamsan et al. [10] presented a unique deep learning (DL) method that combines NLI approaches with GloVe and BERT embeddings. They identified particular violations with up to 76% accuracy, offering a potential path for privacy compliance. The method's restricted emphasis on GDPR Article 9 and its poor validation accuracy of 66%, however, suggest that it needs to be applied more broadly and better aligned with a variety of legal frameworks.

In their review of Named Entity Linking (NEL), NER and

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TABLE I
SUMMARY OF PRIVACY POLICY ANALYSIS IN THE LITERATURE RELATED TO THE PROPOSED WORK

Author	Techniques Used	Advantages	Limitations
Cui [1]	• Introduced PoliGraph, a knowledge graph architecture. • Developed PoliGraph-ER (NLP tool) to extract relations from PPs.	• Enables comprehensive analysis of PPs. • Captures statements as relational data.	• Primary focus is on data collection. • May need enhancements to cover other policy aspects.
Elluri [5]	• Used Semantic Web technologies (OWL, RDF), NLP, and text mining to map GDPR and PCI DSS regulations. • Created an integrated knowledge graph.	• Provides a machine-readable, semantically rich framework. • Automates GDPR and PCI DSS compliance. • Verified against policies of major cloud service providers.	• Coverage limited to GDPR and PCI DSS. • Alignment process is manual. • More extensive automation features are under development.
Bhatia [6]	• Employed a hybrid approach combining NLP and crowd-sourcing. • Extracted privacy implications from PPs.	• High recall (0.85) and precision (0.93) in determining privacy goals. • Effectively combines automation with human judgment.	• Accurate annotations required for effective dependency parsing. • Challenges with complete automation and scalability remain.
Martinelli [7]	• Used transfer learning, unsupervised learning, and NLP. • Identified sensitive information in text-based documents.	• Semi-automated detection of sensitive data. • Reduces manual labor, improved precision and recall rates.	• Challenges in achieving domain-specific knowledge representation. • Validation process still heavily relies on human expert input.
Silva [8]	• Combined NER models with NLP tools including NLTK, Stanford CoreNLP, and spaCy. • Focused on the detection of PII.	• High recall and precision, with up to approx. 90% F1 Score. • Enhances adherence to privacy laws.	• Manual dataset labeling requires significant human effort. • Scalability is limited by the scarcity of high-quality labeled datasets.
Carson [9]	• Employed machine learning models such as logistic regression, Naïve Bayes, SVM, and neural networks. • Utilized NLP techniques like TF-IDF and Bag of Words.	• Automatic classification of PPs, up to 84% accuracy for online policies. • Identified vocabulary overlaps between Internet of Things (IoT) and web policies.	• Vocabulary variances and a smaller sample size resulted in lower accuracy (67%) on IoT PPs. • Inconsistent annotations impacted the accuracy of the results.
Alshamsan [10]	• Deep learning, word embedding models (GloVe and BERT). • Combined Natural Language Inference (NLI) to identify GDPR infractions in PPs.	• Introduced a novel technique using the MultiNLI dataset with the government genre. • Achieved up to 76% accuracy in detecting GDPR violations.	• Results are contingent on dataset alignment. • Focus is narrow, specifically on GDPR Article 8. • Modest Validation accuracy at 66%.
Al-Moslmi [11]	• Examined methods for creating knowledge graphs, including NEL, NER, and NED.	• Focuses on the potential of end-to-end systems. • Describes integrated and deep learning-based methods for entity extraction and linkage.	• Limited multilingual support for NER/NEL tasks. • No standard assessment techniques. • High production costs to create domain-specific datasets.
Andow [12]	• Created PolicyLint, a tool that combines logical contradiction analysis, ontology development, and sentence-level NLP. • Finds inconsistencies in app PPs.	• Automatically detects contradictions in PPs. • Scalably examined 11,430 policies, revealing that 14.2% contain logical contradictions.	• Requires manual confirmation of detected conflicts. • Relies on trained NER models and precise NLP parsing.
Zimmeck [13]	• Developed "Privee", a hybrid approach for evaluating online PPs. • Combines machine learning (rule-based and Naïve Bayes classifiers) with crowd-sourcing.	• Achieved high F-1 scores (90% overall). • Supports real-time policy analysis. • Facilitates privacy transparency through browser extensions.	• Accuracy declines with extremely ambiguous material. • Depends quality of crowd-sourced data. • Constrained by ambiguity in NL policies.

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Named Entry Disambiguation (NED) methods for knowledge graph construction, Al-Moslmi et al. [11] demonstrated the potential of end-to-end systems by highlighting the move towards integrated, DL-based entity extraction and linkage techniques. However, the high costs of domain-specific datasets, the absence of multilingual support, and standard evaluation methods continue to be major obstacles to widespread adoption.

Andow et al. [12] created PolicyLink, a program that uses sentence-level NLP ontology creation, and logical contradiction analysis to find discrepancies in app PPs. The tool's usefulness was demonstrated by its scalable study of over 11,000 policies which showed that 14.2% included inconsistencies. However, the need for trained NER models for accurate NLP parsing and the manual confirmations of conflicts point to areas that need improvement. Zimmeck et al. [13] presented Privee, a hybrid method for assessing online PPs that combines crowd-sourcing and machine learning, including rule-based and Naïve Bayes classifiers. Privee achieved strong F1 scores (90%) and used browser extensions to provide transparency and real-time policy analysis. Notwithstanding these benefits, the method's effectiveness deteriorated when it came across unclear content, underscoring the need for high-quality crowd-sourced input.

A framework for automated PP analysis using Convolutional Neural Networks (CNNs), hierarchical multi-label classification, and custom word embeddings was introduced by Harkous et al. [14]. Polisis demonstrated notable progress in user-centric privacy tools with an 88.4% predication accuracy for privacy icons and an 82% top-3 response accuracy real-time QA system. Its dependence on the OPP-115 taxonomy, high training data needs, and possible susceptibilities to maliciously designed regulations, however, point to areas that deserve more investigation.

Blondel et al. [15] proposed the Louvain method, a poplar community discovery approach that maximizes modularity in large networks. In order to create a new network representation, the approach first repeatedly assigns each node to the community that maximizes modularity gain. Next, it combines the detected communities into super-nodes. Until no further modularity improvement is feasible, this hierarchical procedure is repeated. Large-scale PPKG analysis can benefit from the computationally efficient Louvain technique, which can handle networks with millions of nodes.

Lastly, PurPliance, a tool for automated consistency analysis of PPs, was created by Bui et al. [16] utilizing a hierarchical taxonomy, dataflow analysis, and neural Semantic Role Labeling (SRL). In comparison to state-of-the-art methods, their method showed increased recall (from 10% to 50%) and accuracy (from 19% to 95%). PurPliance might be improved, especially in handling certificate-pinned communications, given its reliance on domain adaption, SRL correctness, and manual confirmation for ambiguous cases.

Issues like scalability, dependence on annotated datasets, and adaption to other domains still exist even though a variety of tools and techniques have been created to evaluate PPs, extract knowledge, and identify discrepancies. In order to develop privacy analysis frameworks that are more reliable, scalable, and easy to use, future research must concentrate on resolving these issues.

III. Methodology

This section describes how knowledge graphs are used to assess PPs from Android e-commerce applications. We start by outlining the methods used to get URLs to PPs from corporate websites via app listings on the Google Play Store, which were later used in PoliGraph (as discussed in Section II) to scrape the PPs. We next describe how we used the PoliGraph-er system to generate knowledge graphs using n-gram models. Then, in order to capture more subtle linkages, we provide our unique modifications to PoliGraph-ER for the creation of skip-gram based knowledge graphs found here. Lastly, we compare the knowledge graphs produced, emphasizing the variations between the two methods. These actions together serve as the cornerstone of our inquiry into the PPs revealed in e-commerce applicationsFig. 1 .

A. Privacy Policy URL Scraping

We gathered URLs linking to the PPs of Android e-commerce apps in order to make the creation of knowledge graphs easier. We scraped the Google Play Store for the top 100 shopping apps using Python, then saved the URLS to their PPs. The Input Layer component in Fig. 2 illustrates the process of scraping and collecting these URLs, which streamlined the generation of knowledge graphs. Accepting URLs as arguments is supported both by our modified fork and the original PoliGraph-er system. The specified PP is automatically scraped from the given URL and directly fed into the PoliGraph-ER pipeline thanks to this functionality. We ensured compatibility with PoliGraph-ER's automated processing capabilities and expedited the policy retrieval process by utilizing these URLs.

B. N-Gram Based Knowledge Graph Generation

As discussed in Section II, the PoliGraph framework captures links between data types, entities, and goals specified in the language of PPs by representing them as knowledge graphs. It makes it possible to link statements from different parts of a policy, facilitating a thorough study that goes beyond interpretations at the phrase level. The framework's implementation, PoliGraph-ER, automatically extracts the connections by utilizing NLP methods, including dependency parsing, NER, and n-gram extraction. In order to find phrases and extract relevant entities and terms for building the knowledge graphs, PoliGraph by default uses n-grams.

PoliGraph represents relationships in PPs using two edge labels: "COLLECT" and "SUBSUME".

- "SUBSUME" indicates hierarchical relationships, such as "Personal information subsumes email addresses and phone numbers".
- "COLLECT" represents data collection, such as a company stating "We collect email addresses".

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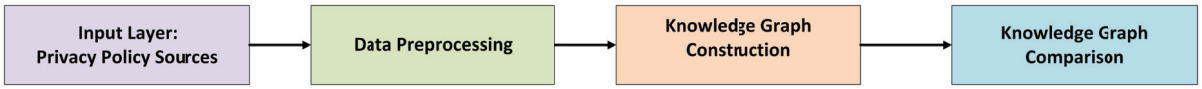

Fig. 1. An overview of the framework for processing PPs into knowledge graphs known as PPKG. URLs scraped from the Google Play Store are gathered by the input layer. The data preprocessing layer processes these URLs by extracting the PP from HTML then feeds the text into two concurrent pipelines, the n-gram NLP pipeline and the skip-gram pipeline. The knowledge graph construction layer receives the results from this and uses graph building to create separate graph databases for skip-gram and n-gram knowledge graphs. In order to assess differences and evaluate improvements in relationships acquired by the skip-gram technique, these knowledge graphs are finally compared in the knowledge graph comparison layer.

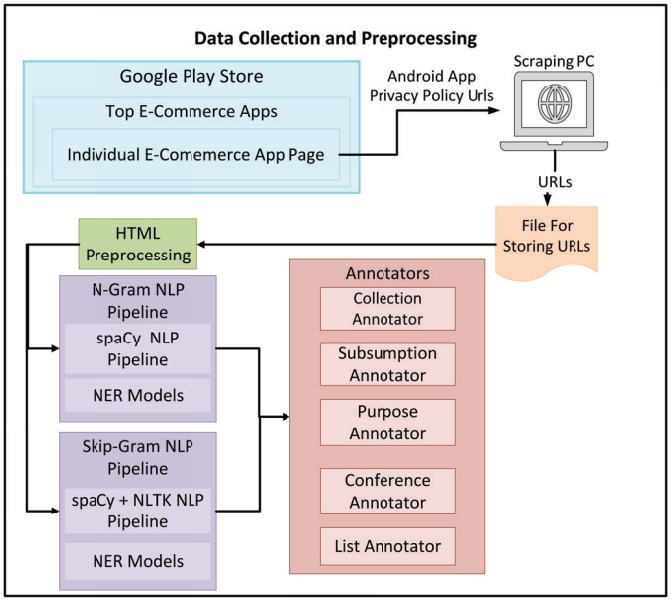

Fig. 2. An overview of the first half of the PPKG framework. URLs scraped from the Google Play Store are processed by extracting PPs from HTML. The text is then fed into two parallel NLP pipelines: n-gram analysis and skip-gram modeling for further processing

Cui et al. developed these terminologies in PoliGraph, which allows PPs to be defined and allows for a more thorough examination of data-collecting methods.

The NLP pipeline, built using the spaCy library and a RoBERTa-based model, analyzes PP content extracted via the scraped URLs. It labels text with linguistic features such as word lemmas, part-of-speech tagging, sentence segmentation, and syntactic dependency parsing. These linguistic features facilitate the identification of key data elements, such as data types and entities, without requiring extensive domain adaptation. NER, a widely used NLP technique, is employed in the pipeline to classify noun phrases into two predefined categories: `DATA`, representing data types, and `ENTITY`, representing organizations and third-party associations.

We used a script in our work to include PoliGraph in our pipeline for analysis. Our script automatically fed the previously scraped PP URLs into the PoliGraph system. After that, the system produced GraphML-formatted knowledge graph representations of privacy regulations. This format was chosen to enable seamless interaction with graph analysis libraries, supporting robust querying and study of the generated knowledge graphs. These knowledge graphs served as the baseline for comparison with the ones produced using our custom modification.

C. Skip-Gram Based Knowledge Graph Generation

As an enhancement to PoliGraph-ER (discussed in Section II), we created a modified version that utilizes skip-gram instead of n-gram. By allowing the addition of non-contiguous word sequences, skip-grams, a generalization of n-grams, capture broader contextual linkages within the text. Due to stringent adjacency requirements, n-grams may miss important connections; this method was created to overcome these restrictions.

In terms of computing, n-gram models typically have a linear time complexity in relation to the corpus's token count. In particular, the complexity of an n-gram model may be written as $O(N \times T)$, where T is the total number of tokens in the corpus and N is the length of the n-gram [17]. This shows that both the size of the n-gram and the number of tokens cause the processing time to grow linearly.

On the other hand, the skip-gram model has a time complexity that varies with the size of the vocabulary and the number of tokens, especially when it is implemented using techniques like Word2Vec. With V denoting the vocabulary size and T denoting the total number of tokens, the complexity is around $O(V \times T)$. According to this, skip-gram models may need more processing power even though they are able to capture more contextual information, particularly when dealing with large vocabularies [18].

Our approach included skip-gram processing in the phrase extraction stage of the PoliGraph-ER pipeline. To find semantically related but non-adjacent phrases in PP language, we specifically modified the sequence matching methods and extended dependency parsing. This allows our system to connect entities and terms between phrases or sentences, producing more thorough and insightful knowledge graph representations. An overview of the KG creation and comparison stage of the PPKG framework is shown in Fig. 3

We tested our skip-gram-enhanced PoliGraph-ER using the same set of PP URLs. The created knowledge graphs were exported in GraphML format to guarantee interoperability with graph analysis tools, just like the baseline n-gram version. Our main goal was to determine if this method may result in higher quality and more relationships being found, especially those involving entities and their purposes. Our goal was to improve the knowledge graphs' analytical capabilities by capturing these subtle relationships, which would enable a more in-depth understanding of the PPs revealed by e-commerce apps.

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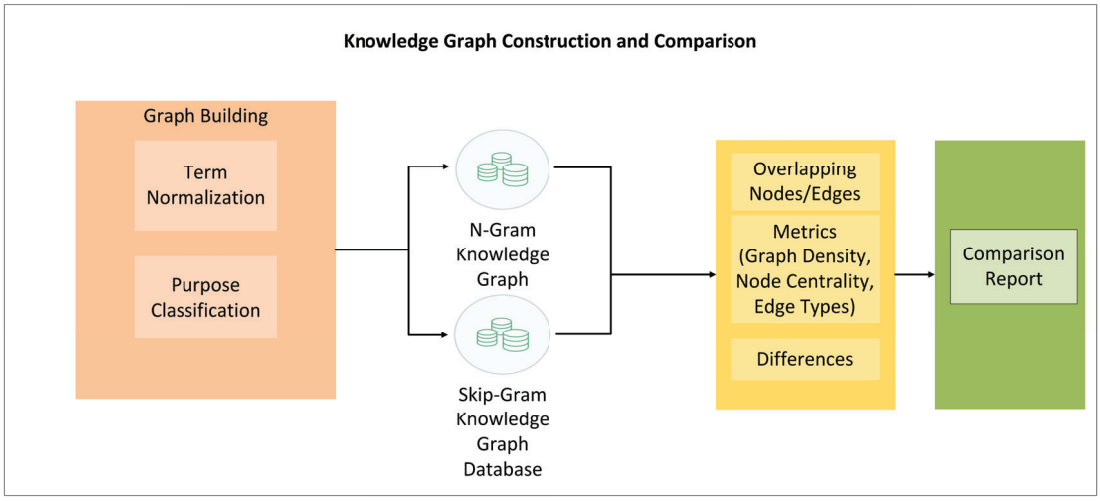

Fig. 3. An overview of the KG creation and comparison stages of the PPKG framework. Outputs from the n-gram and skip-gram pipelines are used to construct separate knowledge graphs, which are then compared to assess differences and evaluate the effectiveness of the skip-gram technique in capturing relationships.

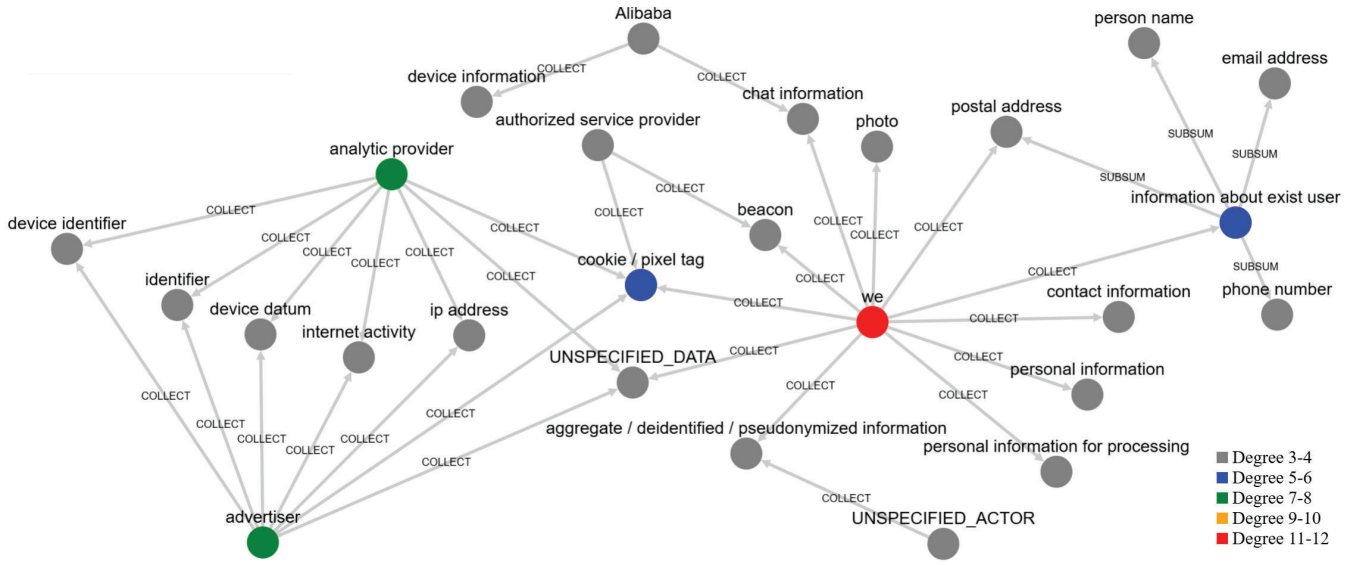

Fig. 4. Data Collection and Sharing in Alibaba's Privacy Policy Graph

D. Comparative Analysis of Knowledge Graphs

Given the KGs, we performed an analysis using graph mining techniques to elucidate the PP frameworks of two major global retailers, Alibaba and Walmart. By visualizing and analyzing the structure of data collection, sharing, and the relationship dynamics within these policies, we aim to uncover insights into each company's approach to data privacy and compliance.

Graph visualization begins with the extraction of structured data from the 'text' component of the PPKGs. This extraction involves identifying key terms and relationships, such as data types collected (e.g., personal information, biometric data), processing actions (e.g., collect, store, share), and involved parties (e.g., third-party service providers). These elements are then transformed into nodes and edges within a graph, where nodes represent entities or data types, and edges depict actions or relationships between these nodes. This step is crucial as it converts verbose legal text into a visual and analytical model that can be systematically analyzed. It allows for the identification of patterns, key actors, and data flows within the privacy policy, making complex information more accessible and understandable. This is particularly important for stakeholders who need to ensure compliance with data protection laws and for end-users seeking transparency about how their data is handled.

Fig. 4 presents the network graph of Alibaba's privacy policy, highlighting nodes like "personal information," "internet activity," and "UNSPECIFIED-DATA." This visualization

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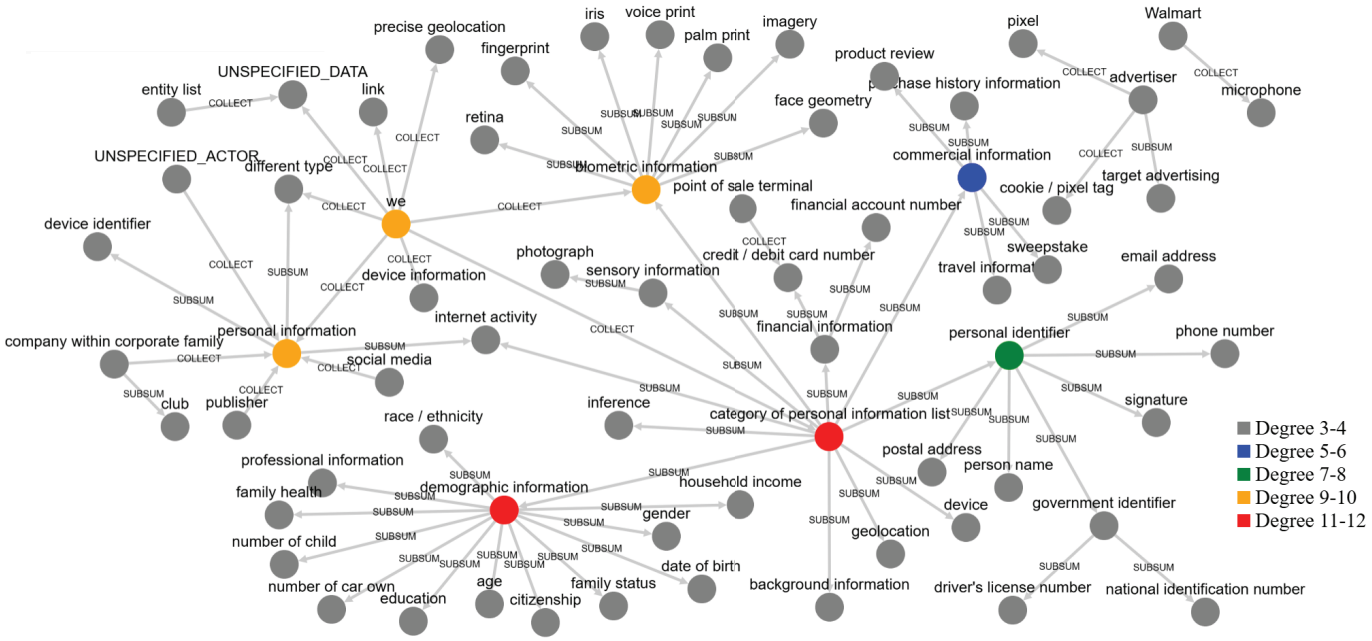

Fig. 5. Data Collection and Sharing in Walmart's Privacy Policy Graph

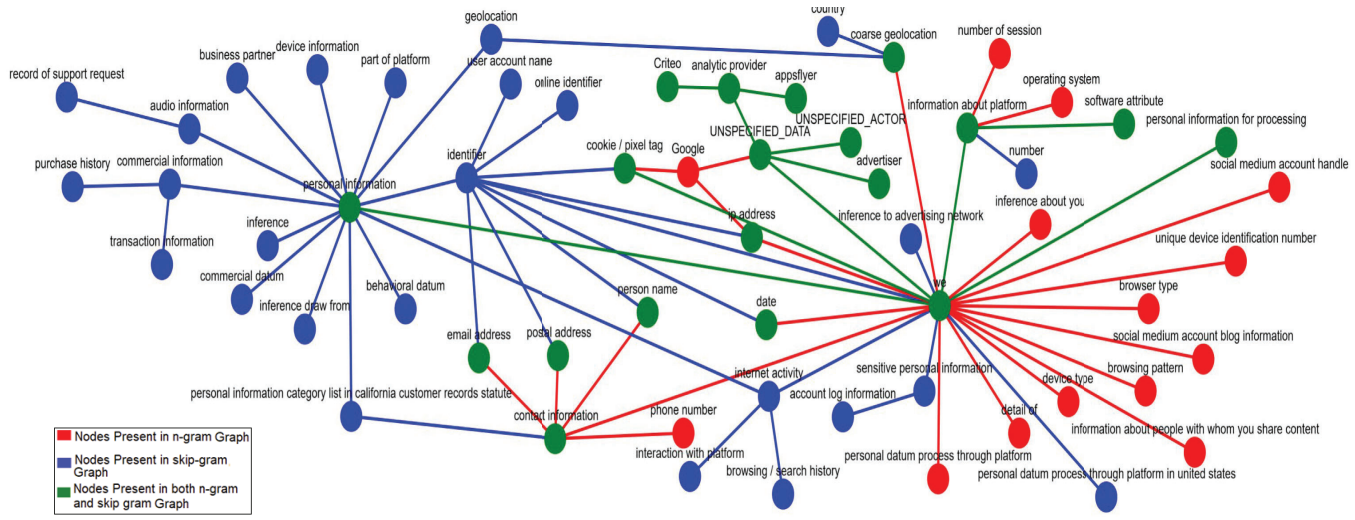

Fig. 6. Comparison of node inclusion between skip-gram and n-gram construction of knowledge graphs in AliExpress' Privacy Policy Graphs

exposes the methods Alibaba uses to gather and distribute user data, focusing on critical nodes that are primary hubs for data collection and dissemination. Fig. 5 illustrates the network graph of Walmart's privacy policy, emphasizing central nodes such as "personal information," "device information," and "biometric information." Nodes are color-coded to represent varying degrees of connectivity; darker colors signify nodes with greater connectivity, illustrating their pivotal role in Walmart's data-handling practices. These graphs provide insights into the key types of data the companies prioritize and their significance in the overall data privacy framework, as well as how extensively data is shared with third parties. Graph Analysis Techniques used for the analysis include *Node Degree Analysis:* calculating the number of connections (or 'degree') each node has. Nodes with higher degrees are typically central to the policy's data practices, indicating key data collectors or frequent data types, *Community Detection (Louvain Method):* [15] to detect communities within the network. A community in this context refers to a group of nodes more densely connected to each other than to other parts of the network, often indicating related policy sections or similar data handling practices, and *Visual Representation:* using tools like Cytoscape to create interactive visualizations of the networks providing in the intuitive understanding of

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complex relationships and hierarchies within the data.

The n-gram and skip-gram models produced similar knowledge graphs. However, skip-gram captured more contextual relationships because it does not strictly rely on adjacent terms. The nodes generated by skip-gram mostly resembled those by n-gram; the main distinction was that skip-gram's larger window size allowed it to catch a wider variety of words. In contrast to n-gram, skip-gram occasionally produced both additional and different entities.

This was especially clear in Figure 6, which shows a KG with notable discrepancies. The blue `graphmlData_skipgram` nodes were created using skip-gram, while the red `graphml_Data` nodes were created using n-gram, highlighting the cases where skip-gram not only detected distinct entities but also extracted a higher number of entities (RQ2). It is crucial to remember that these differences were not the norm for all KGs; rather, they were an anomaly. In the future, we intend to improve entity extraction and KG quality by further optimizing the skip-gram implementation and investigating other contemporary NLP approaches, as covered in Section V.

IV. Discussions

Understanding the structure and relationships within PPKGs provides valuable insights into how companies handle user data. This section discusses key findings from the comparative analysis of PPs, the limitations of the skip-gram model, and their implications for PP analysis.

A. Comparative Analysis and Observations

The graph-based community detection results presented above provide a novel method for uncovering and comparing how different companies approach privacy and data protection. By examining PPKGs, we identified key patterns in data collection, storage, and sharing.

Findings from Alibaba's PP Graph Analysis:

- Centralized data collection is evident, with the node "we" being highly connected, suggesting Alibaba directly collects large amounts of user data.
- The company explicitly collects biometric data, raising privacy concerns due to the sensitive nature of this information.
- The presence of extensive third-party sharing indicates complex data-sharing arrangements, which may impact user privacy and security.

Findings from Walmart's PP Graph Analysis:

- Walmart's graph shows multiple central nodes, indicating a more distributed approach to data collection rather than a single dominant entity.
- Financial and health-related data are prominently collected, requiring strong security and compliance measures to prevent misuse.
- The graph displays fewer third-party connections, which could suggest either a reduced amount of third-party data sharing or a lack of transparency in disclosures.

These findings highlight the importance of well-defined data categories and clear policy disclosures (RQ1). Unspecified data categories and complex third-party interactions may pose challenges in ensuring compliance with regulations like the GDPR. Improving transparency in third-party sharing agreements and refining policy disclosures could help users better understand how their data is handled.

B. Understanding Skip-Gram's Limitations in PP Analysis

Although both n-gram and skip-gram models generated similar knowledge graphs, skip-gram demonstrated an ability to capture broader contextual relationships. However, it also missed certain nodes that were present in n-gram-based graphs. This discrepancy arises from fundamental differences in how these models process text.

One of the key reasons for this difference is the strict adjacency requirement in n-gram models. Because n-grams only consider words that appear next to each other within a fixed window, they ensure all direct phrase sequences are preserved. This makes them particularly useful for detecting explicit policy statements. In contrast, skip-gram introduces a more flexible window by linking words that may be separated by intermediate terms. While this allows it to capture relationships that span across phrases, it can also lead to missing entities when critical words are not co-occurring frequently enough.

Another contributing factor is how these models prioritize word relationships. Skip-gram models rely on statistical co-occurrence, meaning words that frequently appear together are more likely to be linked. As a result, low-frequency terms—especially those found in technical legal statements—may be omitted. In privacy policies, this can result in missing key entities related to data collection or legal provisions.

C. Implications for Privacy Policy Knowledge Graphs

The differences between these models have significant implications for PP analysis. While skip-gram provides a better understanding of broader semantic relationships, it may fail to capture crucial adjacent legal terms, leading to gaps in policy comprehension. For example, if a privacy statement such as "We collect biometric data" is structured across multiple sentences, the skip-gram model might fail to establish a direct link between the collecting entity and the data type.

To address this issue, a hybrid approach could be beneficial. By leveraging n-grams for precise entity extraction and skip-grams for broader relationship detection, knowledge graphs could offer both accuracy and contextual depth. Additionally, refining the skip-gram implementation through dependency parsing or adjusting the window size could reduce node omissions.

These refinements will be crucial for enhancing automated privacy policy analysis tools. Future research can explore these modifications to improve the balance between contextual flexibility and precise entity recognition, ensuring a more complete representation of PP structures.

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V. Conclusion and Future Work

This work has shown how graph mining and NLP tools can be used to make PPs easier to comprehend and uncover related dangers. We demonstrated the advantages of skip-gram-based knowledge graphs over conventional n-gram-based techniques by utilizing them to capture semantic linkages better and offer a more comprehensive representation of policy content. Our strategy highlights how crucial it is to develop new tools and techniques to increase user accessibility and transparency of PPs.

Our study's findings demonstrate important developments in the examination of PPs. Critical insights into data categories, hierarchical structures, and third-party relationships were revealed by the comprehensive visualization of data collection and sharing processes that PPKG offered. For instance, nodes that represented entities like "We" were found to be high-degree nodes, indicating possible weaknesses because of their vast data links. This demonstrates how our method may help forensic investigations and regulatory compliance by identifying trends and connections in PPs. Visualization of PPKGs and graph-based community detection algorithms underscore the complexity of PPs in large multinational corporations. The graph-based visualization and analysis provide a novel method to uncover and compare the intricacies of how different companies approach privacy and data protection.

Future work will explore several directions to enhance and expand this research. First, to evaluate the generalizability and resilience of the suggested approaches, we aim to broaden our research by including PPs from other industries, such as gaming and healthcare. Using advanced graph neural networks will provide more sophisticated insights into the structure and implications of PPs by facilitating deeper reasoning within knowledge graphs. In conjunction, we will further investigate RQ3 and RQ4, focusing on analyzing potential risks to customer rights and personal data privacy, and exploring methods to design software systems that ensure transparency and accountability in data collection, usage, and sharing. These efforts will help develop real-time policy analysis tools that assist users with software installs and provide prompt, useful feedback on privacy implications.

In the technical enhancement of our tools, we plan to improve our fork of PoliGraph-ER (as described in Section II), specifically with better tuning of skip-gram window size. The skip-gram and n-gram KGs were similar. Further experimentation will explore their differences in detail. We also aim to enhance the accuracy and efficiency of PoliGraph-ER by integrating semantic analysis tools that can discern context and intent within PPs more precisely. Deploying NLP algorithms will enable the extraction of deeper meanings from the text, allowing for more nuanced distinctions between different types of data handling practices. By improving our system's ability to understand and categorize text semantically, we anticipate a significant boost in the performance metrics of both skip-gram and n-gram based Knowledge Graphs. Lastly, to provide end users more control over their data while upholding robust privacy guarantees, we suggest investigating the integration of privacy-preserving technologies like federated learning.

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References

[1] H. Cui, R. Trimananda, A. Markopoulou, and S. Jordan, "{PoliGraph}: Automated privacy policy analysis using knowledge graphs," in *32nd USENIX Security Symposium (USENIX Security 23)*, 2023, pp. 1037–1054.

[2] J. R. Deller and J. Hansen, "3 - methods, models, and algorithms for modern speech processing," in *The Electrical Engineering Handbook*, W.-K. CHEN, Ed. Burlington: Academic Press, 2005, pp. 861–890. [Online]. Available: https://www.sciencedirect.com/science/article/pii/B9780121709600500633

[3] T. Mikolov, I. Sutskever, K. Chen, G. Corrado, and J. Dean, "Distributed representations of words and phrases and their compositionality," *arXiv preprint arXiv:1310.4546*, 2013. [Online]. Available: https://arxiv.org/abs/1310.4546

[4] J. A. Obar and A. Oeldorf-Hirsch, "The biggest lie on the internet: Ignoring the privacy policies and terms of service policies of social networking services," *Information, Communication & Society*, vol. 23, no. 1, pp. 128–147, 2020.

[5] L. Elluri, A. Nagar, and K. P. Joshi, "An integrated knowledge graph to automate gdpr and pci dss compliance," in *2018 IEEE International Conference on Big Data (Big Data)*. IEEE, 2018, pp. 1266–1271.

[6] J. Bhatia, T. D. Breaux, and F. Schaub, "Mining privacy goals from privacy policies using hybridized task recomposition," *ACM Transactions on Software Engineering and Methodology (TOSEM)*, vol. 25, no. 3, pp. 1–24, 2016.

[7] F. Martinelli, F. Marulli, F. Mercaldo, S. Marrone, and A. Santone, "Enhanced privacy and data protection using natural language processing and artificial intelligence," in *2020 International Joint Conference on Neural Networks (IJCNN)*. IEEE, 2020, pp. 1–8.

[8] P. Silva, C. Gonçalves, C. Godinho, N. Antunes, and M. Curado, "Using nlp and machine learning to detect data privacy violations," in *IEEE INFOCOM 2020-IEEE conference on computer communications workshops (INFOCOM WKSHPS)*. IEEE, 2020, pp. 972–977.

[9] J. Carson, L. DiSalvo, and L. Ray, "Automatic classification of web and iot privacy policies," in *2022 IEEE 19th International Conference on Mobile Ad Hoc and Smart Systems (MASS)*. IEEE, 2022, pp. 732–735.

[10] A. R. Alshamsan and S. A. Chaudhry, "Detecting privacy policies violations using natural language inference (nli)," in *2022 IEEE Asia-Pacific Conference on Computer Science and Data Engineering (CSDE)*. IEEE, 2022, pp. 1–6.

[11] T. Al-Moslmi, M. G. Ocaña, A. L. Opdahl, and C. Veres, "Named entity extraction for knowledge graphs: A literature overview," *IEEE Access*, vol. 8, pp. 32 862–32 881, 2020.

[12] B. Andow, S. Y. Mahmud, W. Wang, J. Whitaker, W. Enck, B. Reaves, K. Singh, and T. Xie, "Policylint: Investigating internal privacy policy contradictions on google play," in *28th USENIX Security Symposium (USENIX Security 19)*. USENIX Association, August 2019.

[13] S. Zimmeck and S. M. Bellovin, "Privee: An architecture for automatically analyzing web privacy policies," in *23rd USENIX Security Symposium (USENIX Security 14)*. USENIX Association, August 2014.

[14] H. Harkous, K. Fawaz, R. Lebret, F. Schaub, K. G. Shin, and K. Aberer, "Polisis: Automated analysis and presentation of privacy policies using deep learning," in *27th USENIX Security Symposium (USENIX Security 18)*. USENIX Association, August 2018.

[15] V. D. Blondel, J.-L. Guillaume, R. Lambiotte, and E. Lefebvre, "Fast unfolding of communities in large networks," *Journal of statistical mechanics: theory and experiment*, vol. 2008, no. 10, p. P10008, 2008.

[16] D. Bui, Y. Yao, K. G. Shin, J.-M. Choi, and J. Shin, "Consistency analysis of data-usage purposes in mobile apps," in *Proceedings of the 2021 ACM SIGSAC Conference on Computer and Communications Security (CCS '21)*. Association for Computing Machinery, 2021.

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[17] D. Jurafsky and J. H. Martin, "Speech and language processing: An introduction to natural language processing, computational linguistics, and speech recognition."

[18] S. Bartunov, D. Kondrashkin, A. Osokin, and D. Vetrov, "Breaking sticks and ambiguities with adaptive skip-gram," in *artificial intelligence and statistics*. PMLR, 2016, pp. 130–138.

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